

Pranav Balachander

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EDUCATION

The Pennsylvania State University

Bachelor of Science in Computer Science | Dean's List, Fall 2025

State College, PA

Graduated 2026

EXPERIENCE

Research Assistant

Dept. of Computer Science @ Penn State

Harrisburg, PA

Apr. 2025 – May 2026

- Authored three research papers (submitted, pending publication) as a named author, spanning electricity price forecasting, Arctic mercury modeling, and pesticide-resistance surveillance.
- Developed a hybrid CNN-BiLSTM-Attention model in Python (TensorFlow, Keras) achieving ~25% lower RMSE (12.40 vs 16.49) and higher R^2 (0.87 vs 0.63) than the strongest baseline models on real-world day-ahead electricity price data.
- Built modeling frameworks integrating quantum-chemistry simulations, epidemiological datasets, and genomic resistance markers into spatiotemporal ML pipelines using Pandas, NumPy, and Scikit-learn.

AI/ML Intern

SBS IT Solutions

Dallas, TX (Remote)

Jun. 2025 – Sep. 2025

- Built a RAG-based AI health chatbot from scratch (front-end and backend) that answered user symptom questions, using LangChain, OpenAI APIs, and a vector database for retrieval-augmented responses.
- Applied LLM concepts such as architectures, prompt engineering, embeddings, and vector databases from a Generative AI program to design and implement the end-to-end system.
- Collaborated with the front-end team on a HIPAA-compliant medical application, contributing to UI development and testing.

Data & Analytics Intern

mResult Corporation

Bangalore, India

Jun. 2023 – Oct. 2023

- Organized and analyzed large datasets using data-warehousing techniques to support analytics workflows.
- Performed data cleaning, manipulation, and exploratory analysis using Pandas and Jupyter Notebook.
- Automated data-processing tasks with Dataiku and built dashboards translating insights for non-technical stakeholders.

PUBLICATIONS

Unified Cross-Regional Time-Series Day-Ahead Electricity Price Dataset (UniElecPrice) | IEEE Sep. 2025

Introduced a globally unified dataset of day-ahead electricity prices from 40 markets across 5 continents, enabling standardized forecasting research; developed and validated a hybrid CNN-BiLSTM-Attention model outperforming LSTM and Transformer baselines. Accepted, IEEE Data Descriptions, 2025.

PROJECTS

AI-Driven Modeling of Mercury Concentration in Arctic Environments

Dec. 2025

- Developed a machine learning-assisted computational framework to model mercury reduction, re-emission, and diffusion processes in Arctic snow and ice environments.
- Integrated quantum chemistry calculations and molecular dynamics simulations with data-driven models to capture multi-scale mercury behavior.
- Applied predictive modeling and validation techniques to improve the accuracy of global mercury cycle simulations under changing environmental conditions.

Day-Ahead Electricity Price Forecasting | Python, TensorFlow, Keras, SHAP

Jul. 2025

- Developed a hybrid deep learning model combining CNN, BiLSTM, and Multi-Head Attention to forecast 24-hour electricity prices from 48-hour historical data.
- Implemented Keras Tuner with Bayesian optimization to fine-tune hyperparameters such as LSTM units, dropout rate, and learning rate.

- Achieved superior performance over baseline models (LSTM, Transformer) across MAE, RMSE, SMAPE, and R^2 metrics using real-world market datasets.
- Generated SHAP summary plots and prediction comparison graphs to visualize feature importance and enhance model interpretability.

Pesticide Resistance in Vector Populations

Dec. 2025

- Developed a data-driven surveillance and modeling framework to track and predict pesticide resistance in mosquito populations at a global scale and tick populations within Dauphin County, Pennsylvania.
- Integrated epidemiological datasets, genomic resistance markers, and environmental variables into spatiotemporal machine learning models to identify emerging resistance patterns.
- Evaluated intervention effectiveness and generated actionable insights to support public health decision-making for vector control strategies.

Maargdarshak | Streamlit, Python, OpenAI API, HTML/CSS

Nov. 2024

- Designed and implemented a Streamlit-based user interface optimized for usability and responsiveness to improve accessibility of AI-driven guidance tools.
- Integrated Python logic with OpenAI's API to enhance interactivity and ensure real-time feedback between frontend and backend.
- Collaborated with a cross-functional team to align UI/UX design with product goals, focusing on user experience and performance.
- Incorporated user feedback and iterative testing to continuously refine design and optimize system performance.

Online Restaurant Management System | PHP, MySQL, HTML

Sep. 2023

- Built a dynamic order management platform to replace manual workflows in small restaurants, improving efficiency and user satisfaction.
- Designed an intuitive interface using HTML and PHP, enabling staff to easily manage food categories, orders, and customer records.
- Used MySQL for secure data storage and fast retrieval, ensuring reliable operations and data integrity.
- Gathered informal user feedback and iterated features based on business needs, prioritizing automation and real-time updates.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, JavaScript, HTML/CSS, PHP

Frameworks: React, Node.js, TensorFlow, Keras, PyTorch, Scikit-learn, Streamlit, Flask

Developer Tools: Git, GitHub, VS Code, Visual Studio, PyCharm, Jupyter, LaTeX

Libraries: pandas, NumPy, Matplotlib, SHAP, XGBoost, OpenAI, LangChain, BeautifulSoup

Databases: MySQL, SQLite, PostgreSQL

Cloud & Deployment Tools: Netlify, Tableau, PowerBI, Figma, Trello, Adobe Photoshop, Adobe Illustrator